OOP course

Voice Assistant

Team 03

Table of content

what do we
have here?

Team member

Overview

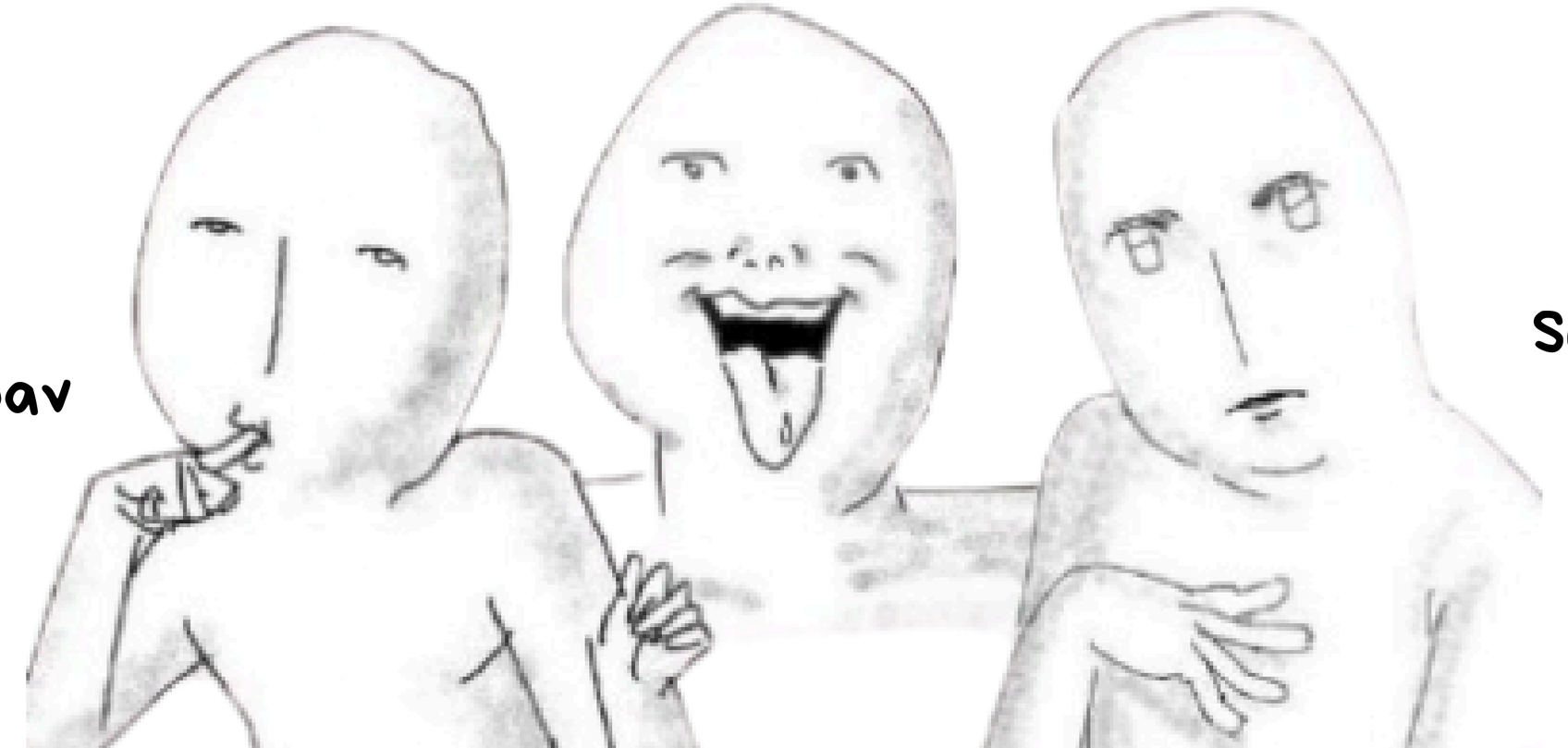
Key feature

Feasibility

Timeline

Team member!!!

Hong Limhak



Keo Seavpav

Vean Sovanvirak

Project Overview

Product

Voice Assistance is a Python text-to-speech (TTS) system that can transfer from text to voice which can overlay on Microsoft Word and other apps for ease of use.

Product important

A lab study tool that can change student behavior from being lazy in reading to being ready to listen. Main focus is student and illiterate people in Cambodia.

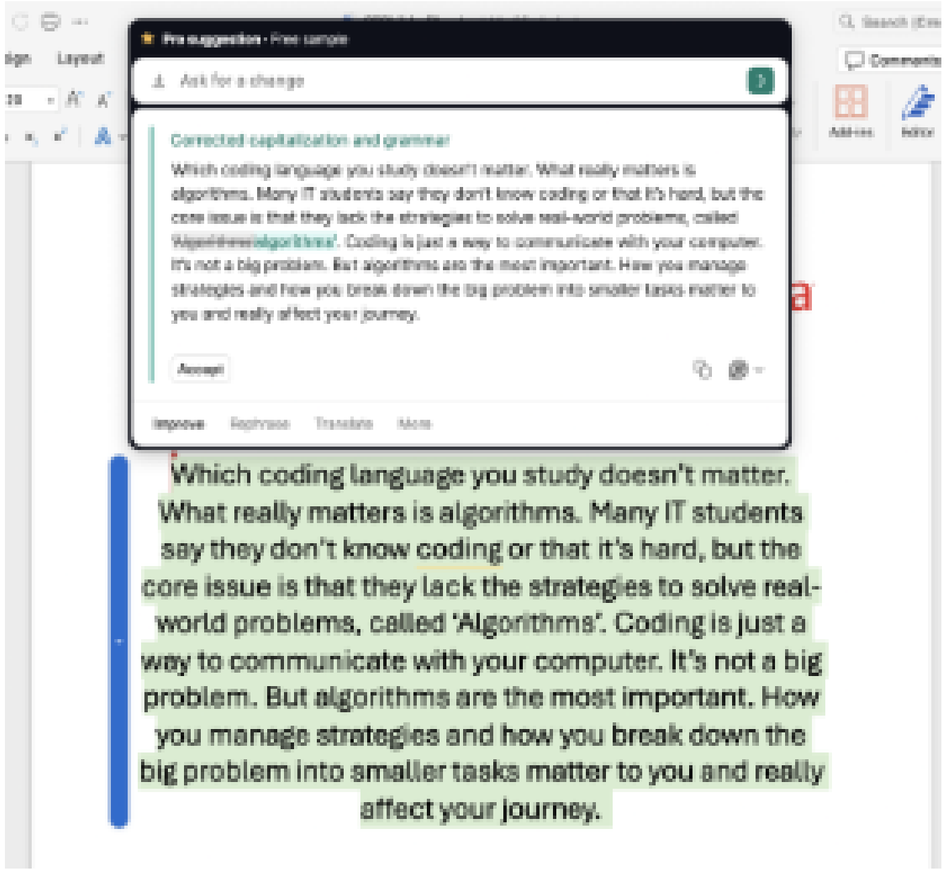
Key features

- Text to Speech (core)
- Background clipboard listener
- History Handling
- GUI

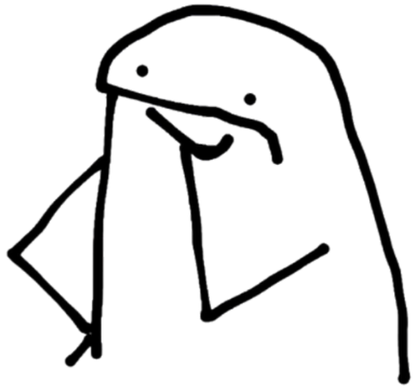
Text-to-speech

- Convert selected text to speech
- using pyttsx3 for offline conversion
- accept manual input
- file and error handling
- Queue based speaking
- related to OOP

Background listener



- Listen to user ctrl+c
- Push item into software
- appear above other software

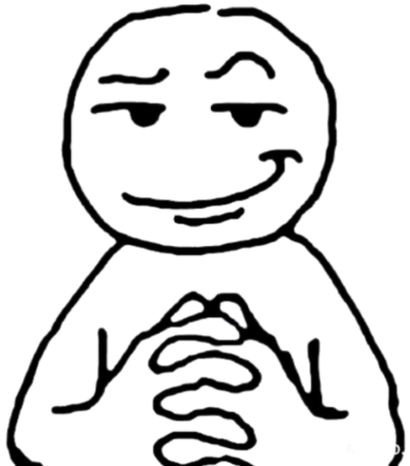


History Handling

```
I do not know whether many people realize how much more
than is ever written there really is in a story—how many
parts of it are never told—how much more really happened
than there is in the book one holds in one's hand and
pores over. Stories are something like letters. When a
letter is written, how often one remembers things omitted
and says, "Ah, why did I not tell them that?" In writing
a book one relates all that one remembers at the time,
and if one told all that really happened perhaps the book
would never end. Between the lines of every story there
is another story, and that is one that is never heard and
can only be guessed at by the people who are good at
guessing. The person who writes the story may never know
all of it, but sometimes he does and wishes he had the
chance to begin again.
12:30:00 12/2/2026

When I wrote the story of "Sara Crewe" I guessed that a
great deal more had happened at Miss Minchin's than I had
had time to find out just then. I knew, of course, that
there must have been chapters full of things going on all
the time; and when I began to make a play out of the book
and called it A Little Princess, I discovered three acts
full of things. What interested me most was that I found
that there had been girls at the school whose names I had
not even known before.
12:40:00 12/3/2026
```

- Push read item to txt
- Have time stamp for each block
- Good for tracking



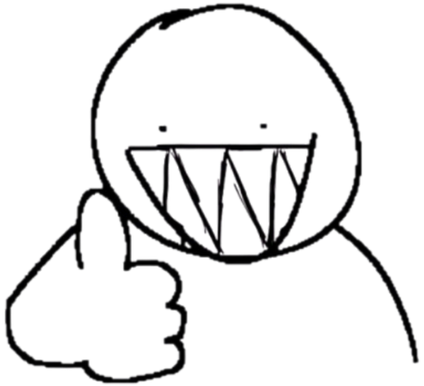
GUI

Link for demostration.

Feasibility

Proven Libraries

pyttsx3 : offline TTS no internet require, Well document, easy to integrate with OOP



PyQt6 : Mature framework for desktop GUI. Qthread prevent UI freeze during speech

keyboard +Pillow+OpenCV: All available via pip.USed in real_world apps - no risk of missing features.

Manageable Scope

Clear Role Separation: Each member owns one module such as TTS Engine, Overlay/Hotkey, GUI-no overlap, no confusion.

OOP Architecture: TTSEngine class encapsulates complexity.
Easy to text independently before integration.

Scoped Feature Set: English only, 250-word limit - intentional constraints that keep scope under control.

Realistic Timeline

Week 1 -Research

Concept understanding for each member's assigned area. Low-risk phase.

Week2+3 - Build & implement

Core coding sprint .Modular design means parallel progress across all 3 members. Combine all code and connect with GUI

Week 4 Integrate

Software testing and Debugging. Code Finalization, GUI Finalization. Wrapping up

Timeline Overview

W1

- Research TTS & pyttsx3
- Explore overlay workflow
- Design Figma UI mockup
- Explore PyQt6 basics

W2

- Build TTS Engine & apply OOP concept
- Code Implementation

W3

- Code Integration + combination with GUI
- Code Implementation

W4

- Debugging and software testing
- Code Finalization
- GUI finalization
- Slide presentation
- Wrapping up

Thank You